

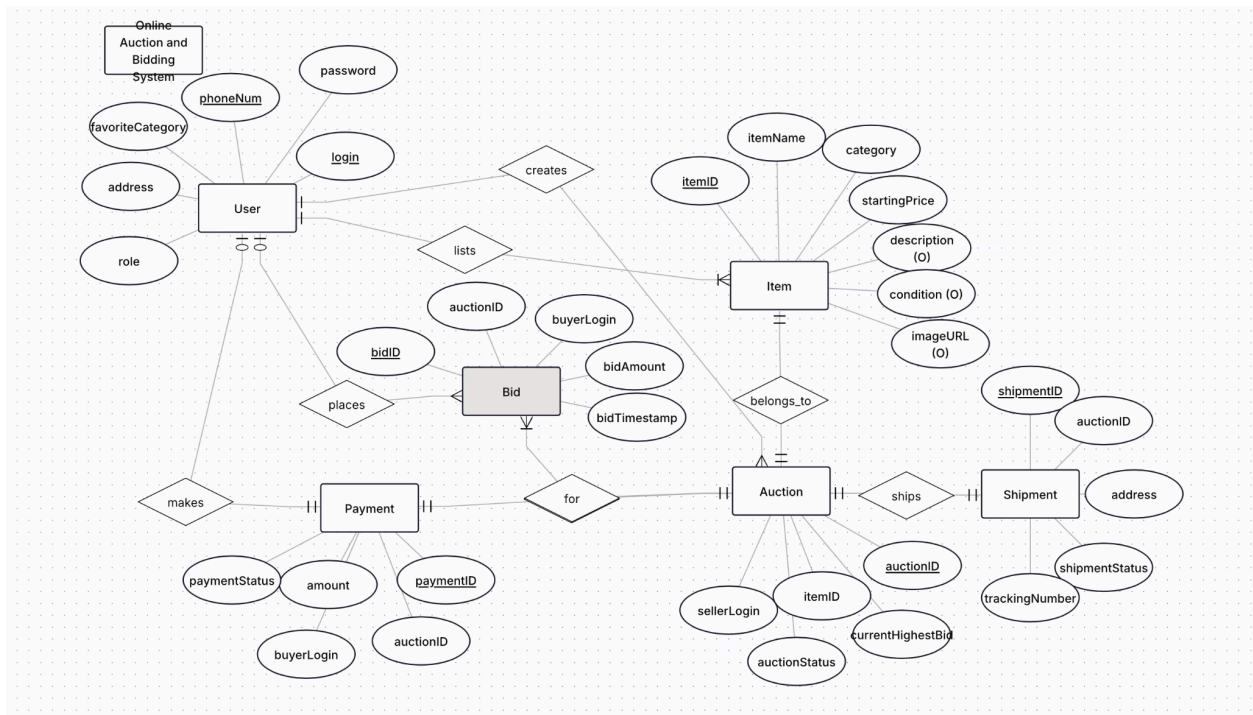
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Phase #1

Diagram:



Supporting Assumptions and documentation:

1. User - Item (lists)
 - a. A seller lists items
 - b. Cardinality:

- i. User (1) -> (M) Item
 - ii. Item -> Exactly 1 user
 - c. Participation:
 - i. Item: Total
 - ii. User: partial
- 2. Item - Auction (belongs_to)
 - a. Cardinality:
 - i. Item (1) -> (1) Auction
 - b. Participation
 - i. Both Total
- 3. User - Auction (Creates)
 - a. Cardinality:
 - i. User (1) -> (M) Auction
 - ii. Auction -> exactly 1 seller
- 4. User - Bid - Auction (Places)
 - a. User - Bid
 - i. User (1) -> (M) Bid
 - ii. Bid-> Exactly 1 user
 - b. Auction - Bid
 - i. Auction (1) -> (M) Bid
 - ii. Bid -> Exactly 1 Auction
 - c. Participation:
 - i. Bid: Total (both must belong)

- ii. Others: partial

5. Auction - Payment (for)

- a. Cardinality:

- i. Auction (1) -> (0 or 1) Payment

- 1. NOTE: 1 payment if the auction is successful, 0 if not

- b. Participation:

- i. Payment: Total

- ii. Auction: Partial

6. User - Payment (makes)

- a. Cardinality:

- i. User (1) -> (1) Payment

- ii. Payment -> exactly 1 User

- b. Participation:

- i. User: partial; if bid highestBid

- ii. Payment; total, a payment is required

7. Auction - Shipment (ships)

- a. Cardinality:

- i. Auction (1) -> (0 or 1) Shipment